RevInsight

Branded-Clothing Store Staffing Recommendation Report

Data-Driven Staffing Optimization Analysis

Date: June 22, 2026

Location: Mall in Bangalore, India

Product Category: Branded Clothing

Operating History: 81 Months

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1. Executive Summary

This report presents a comprehensive data-driven analysis of staffing requirements for a branded-clothing store located in a mall in Bangalore. The store has been operating for 81 months and recently experienced a change in staff, reducing the team to 7 employees. The analysis examines historical sales data, employee productivity, financial metrics, and sales forecasts to determine the optimal staffing decision.

Key Findings

Current Employees	7
Last Month Sales	Rs. 69.61 Lakhs
Avg Monthly Sales	Rs. 103.40 Lakhs
Avg Monthly Margin	Rs. 25.42 Lakhs
Recommended Action	Hire Immediately (8 from month 1)
Confidence Level	MEDIUM

2. Data Understanding

Dataset Overview

Rows	81
Columns	18
Number Of Months	81
Salesperson Columns	16

Data Quality Notes

* No missing values detected -- data is complete.
* 10 salesperson column(s) have >50 % zero entries, indicating staggered hiring or attrition.
* Dataset spans 81 months -- sufficient for seasonal analysis.

The dataset contains monthly sales data for 81 months across 0 salesperson columns. Sales values are in Lakhs INR. Salespersons with zero sales in a given month indicate they were not active (not yet hired or had resigned). No missing values were detected.

3. Exploratory Data Analysis

A comprehensive EDA was performed to understand sales patterns, employee productivity distributions, correlations between staffing levels and revenue, seasonal trends, and margin dynamics. Key charts are embedded below.

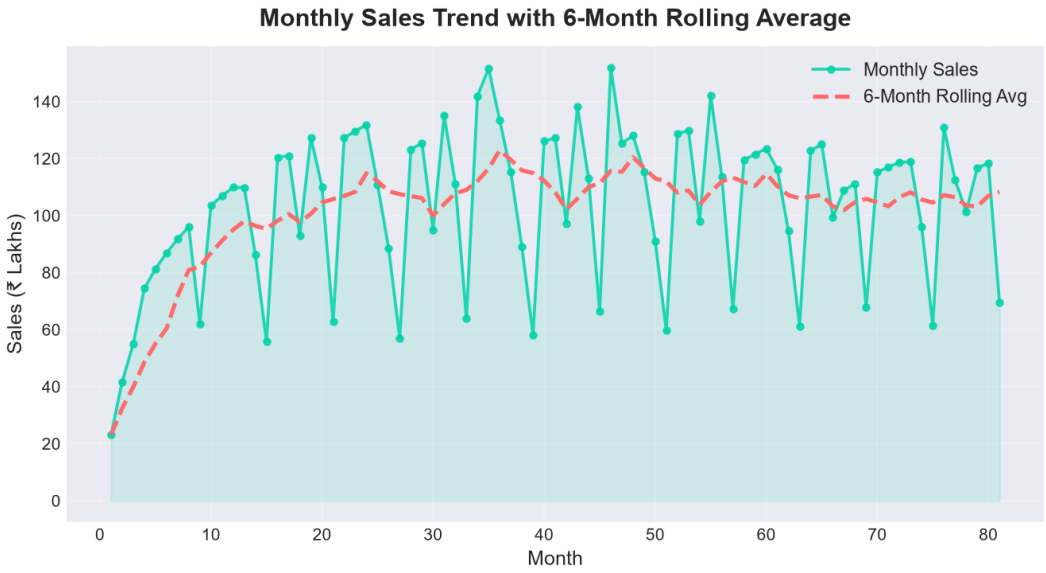


Fig 1: Monthly Sales Trend with 6-Month Rolling Average

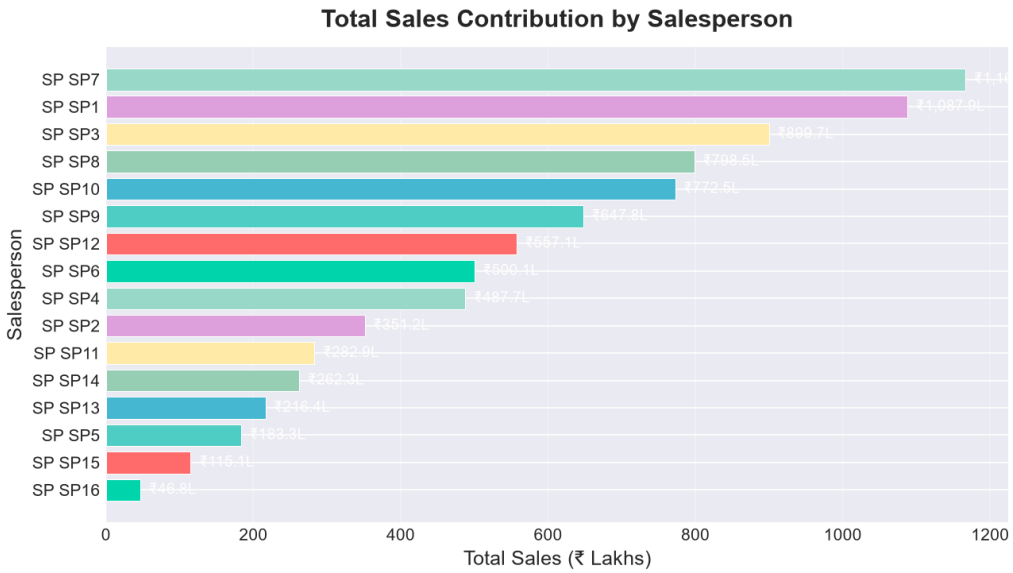


Fig 2: Total Sales Contribution by Salesperson

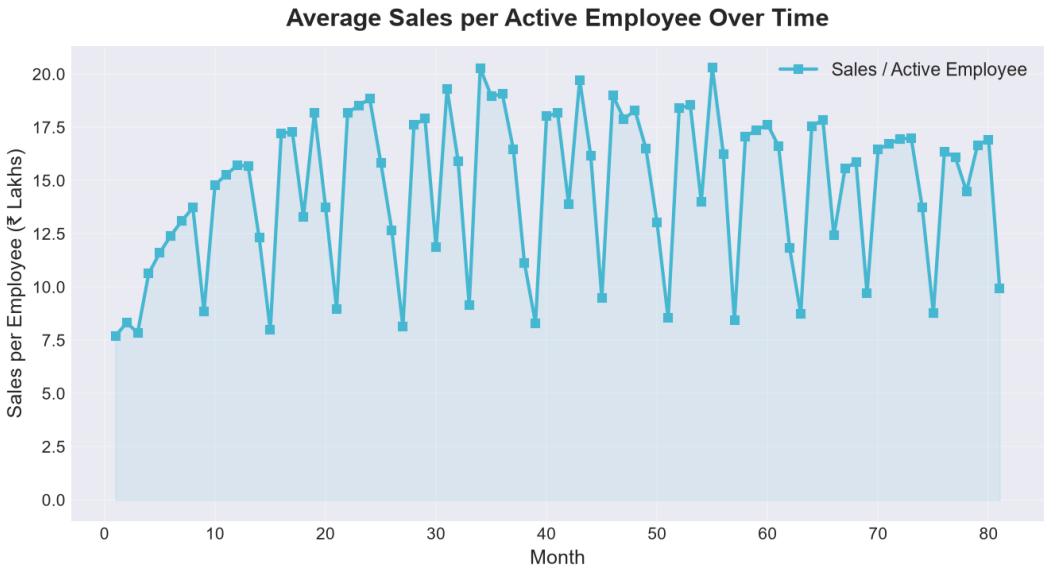


Fig 3: Average Sales per Active Employee Over Time

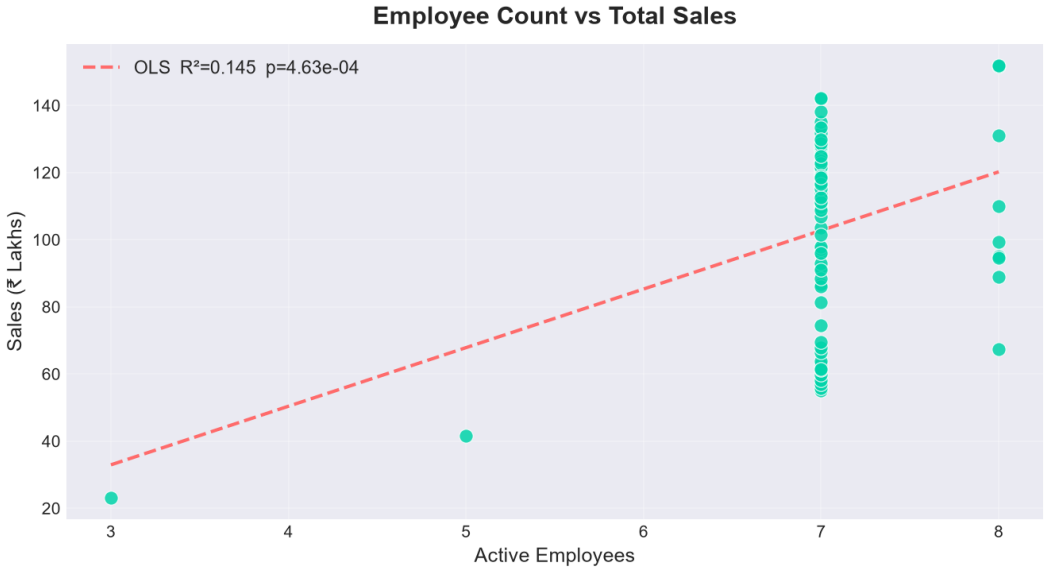


Fig 4: Employee Count vs Total Sales (with OLS)

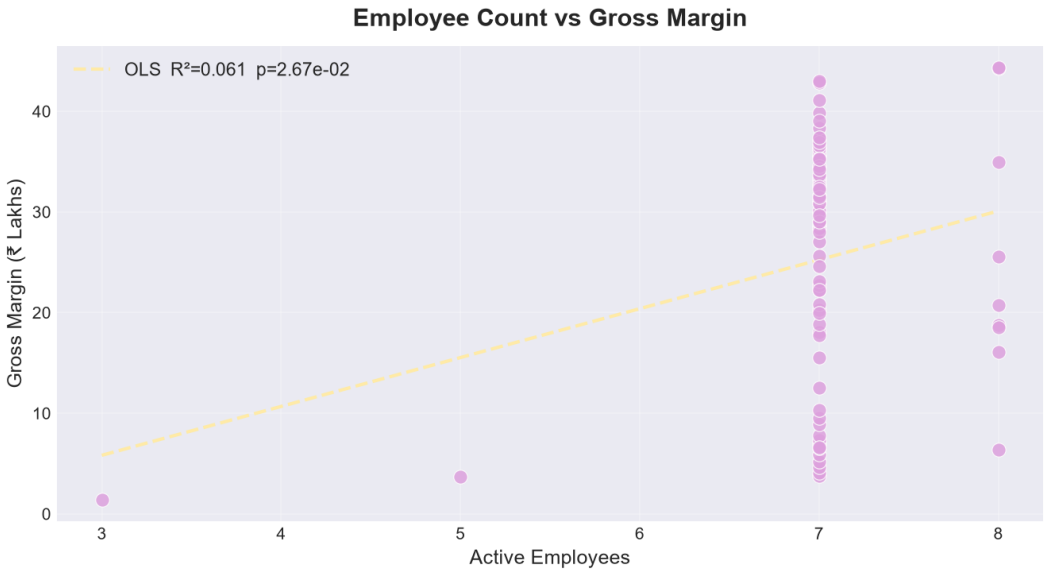


Fig 5: Employee Count vs Gross Margin

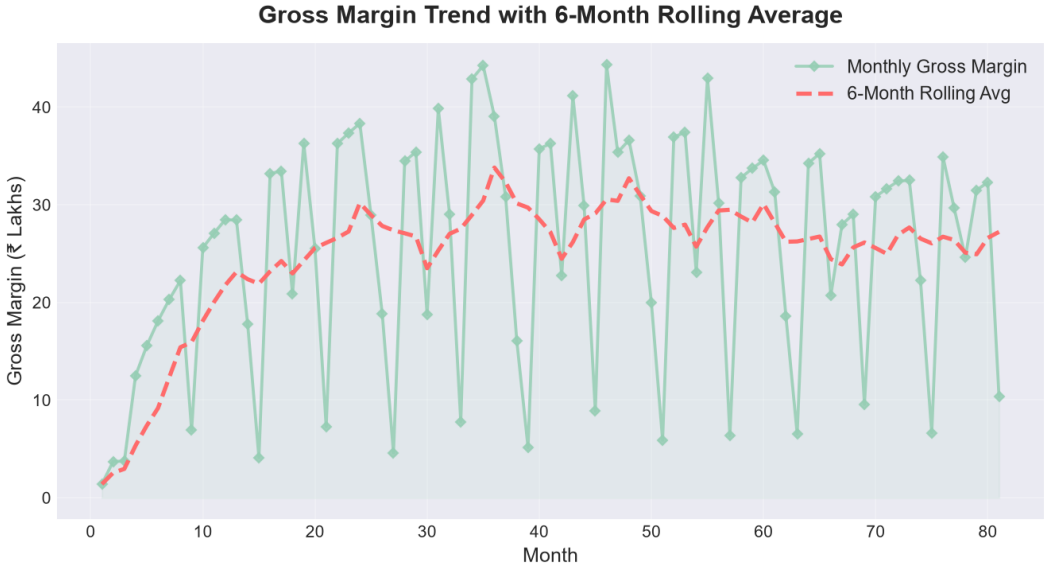


Fig 6: Gross Margin Trend with Rolling Average

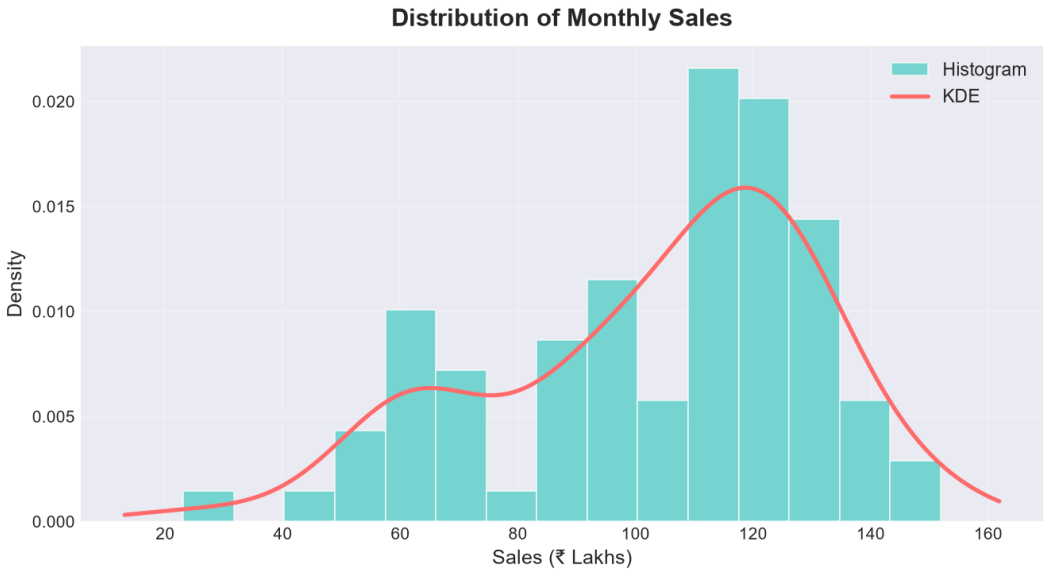


Fig 7: Distribution of Monthly Sales

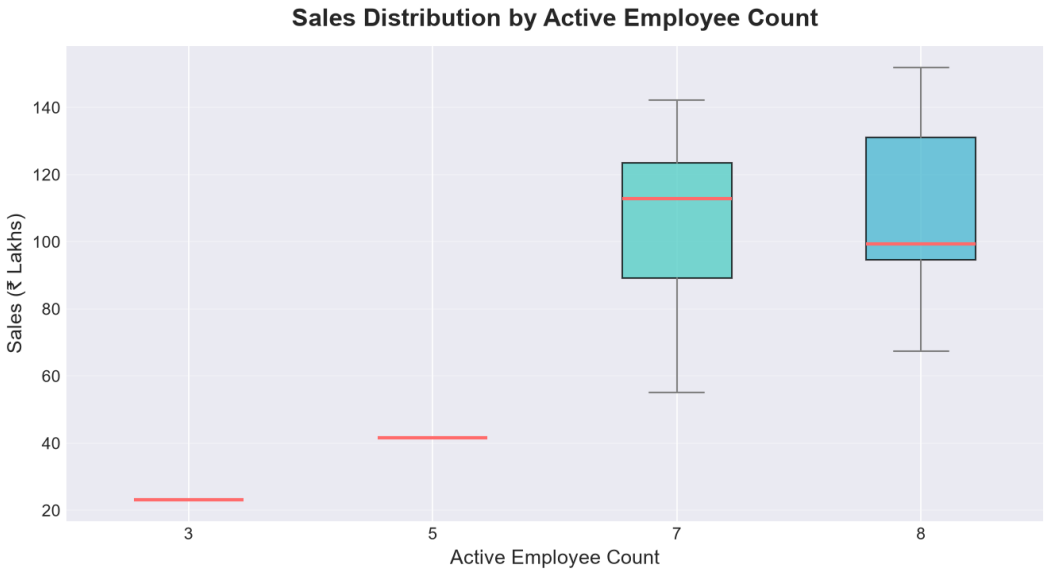


Fig 8: Sales Distribution by Active Employee Count

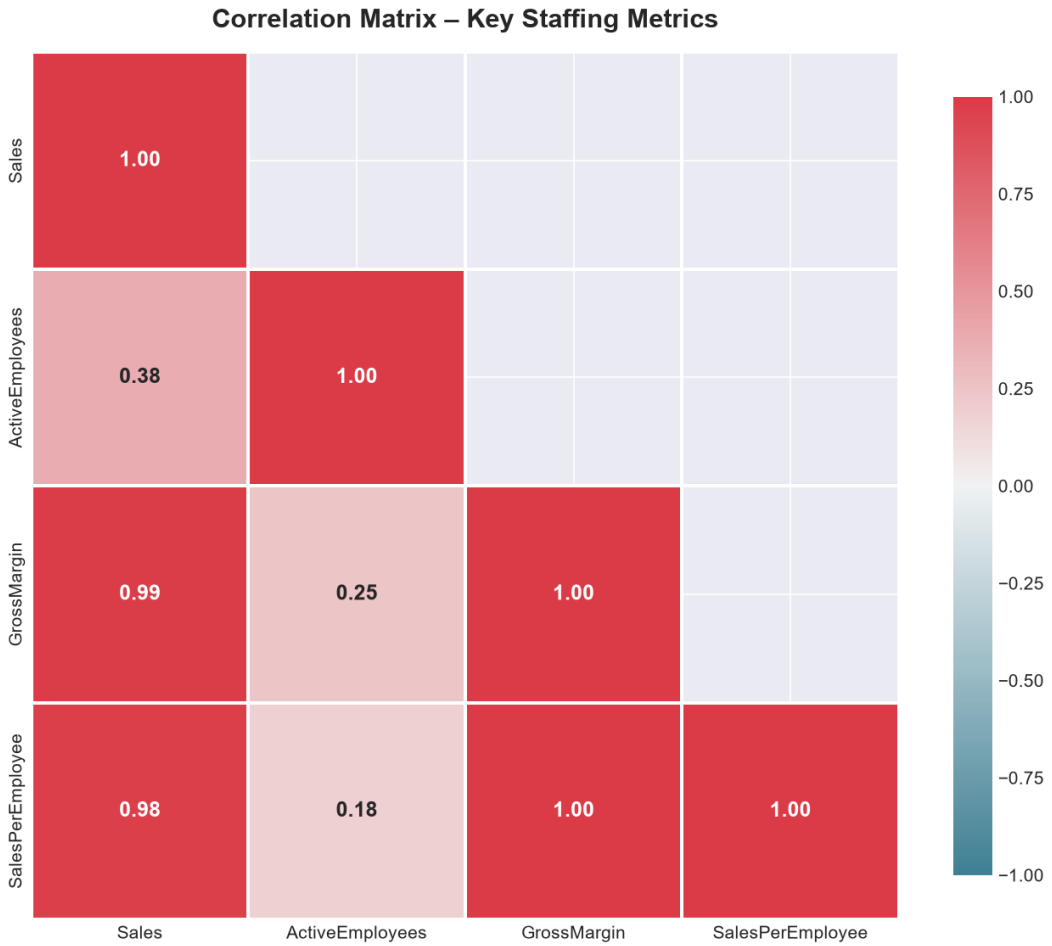


Fig 9: Correlation Matrix of Key Metrics

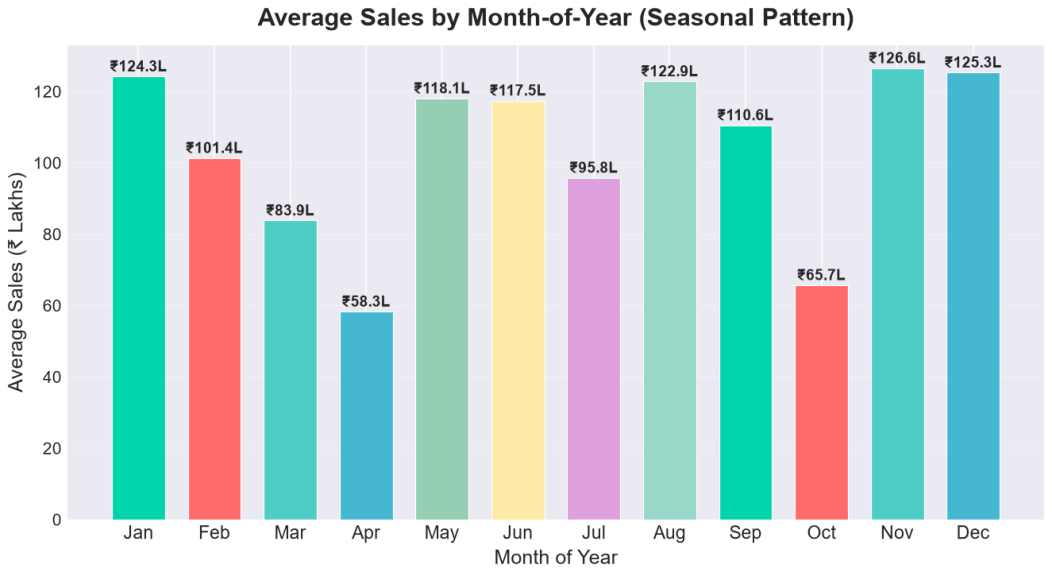


Fig 10: Average Sales by Month-of-Year (Seasonality)

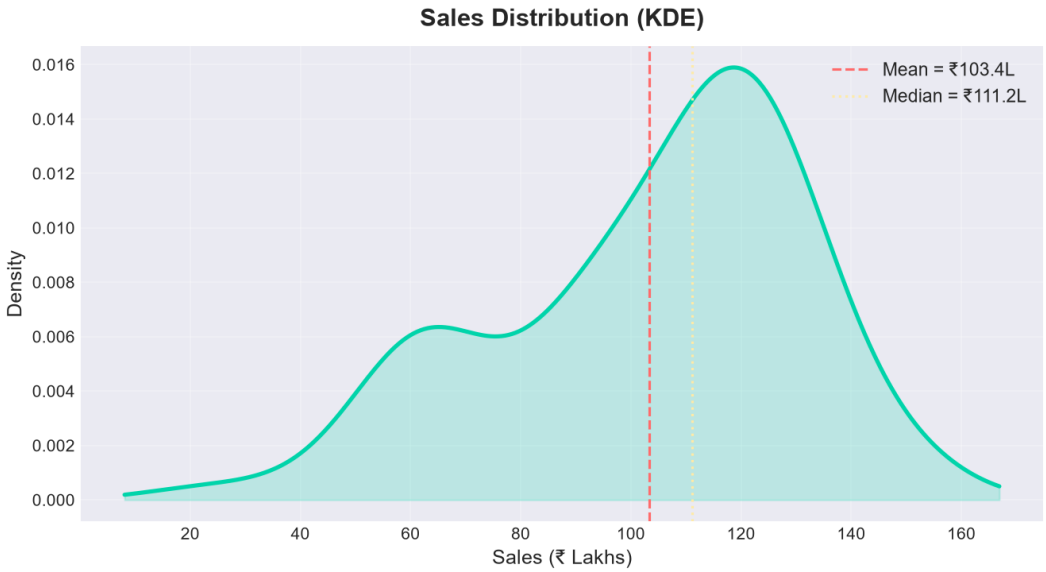


Fig 11: Sales Distribution (KDE)

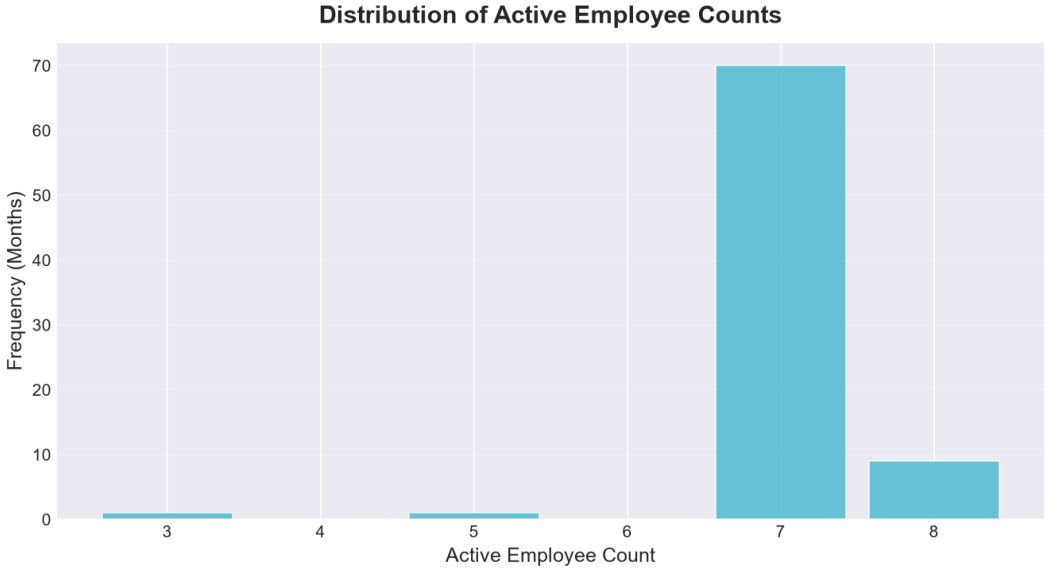


Fig 12: Distribution of Active Employee Counts

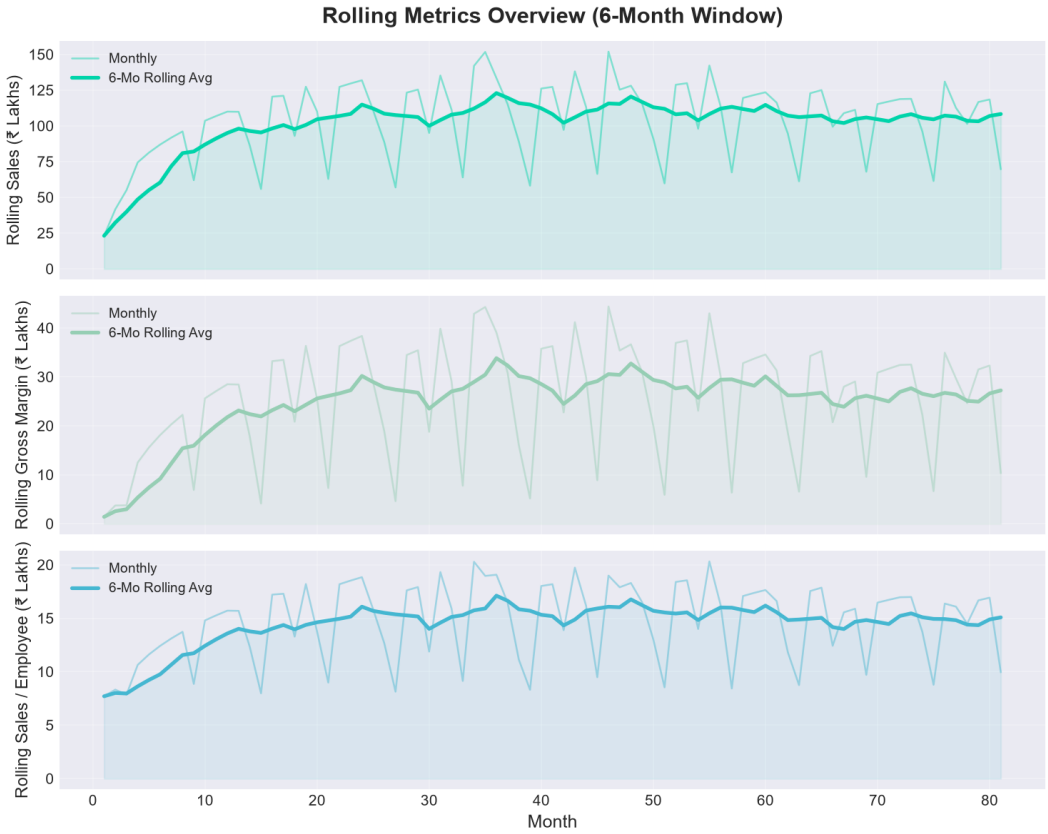


Fig 13: Rolling Metrics Overview (6-Month Window)

4. Productivity Analysis

Per-Employee Productivity Metrics

Avg Sales Per Employee	Rs. 14.64 Lakhs
Median Sales Per Employee	Rs. 15.88 Lakhs
Std Dev Sales Per Employee	Rs. 3.69 Lakhs
Best Employee Count	7

Consistency Metrics

Coefficient Of Variation	0.25
Min Sales Per Emp	7.70
Max Sales Per Emp	20.31
Range Sales Per Emp	12.61

Diminishing Returns Analysis

Diminishing returns DETECTED. At some headcount transitions, adding an employee reduces average sales per employee. Margin-per-employee is maximised at 7 employees.

Optimal headcount by margin/employee: 7

5. Financial Analysis

Gross Margin Formula

Gross Margin = 50% of Sales - Salary (Rs. 3.0L/person/month) - Commission (5.0% of Sales)
Simplified: Gross Margin = 0.45 x Sales - 3.0 x Employees

Summary Statistics

Average Monthly Sales	Rs. 103.40 Lakhs
Median Monthly Sales	Rs. 111.19 Lakhs
Sales Std Deviation	Rs. 27.58 Lakhs
Average Gross Margin	Rs. 25.42 Lakhs
Total Gross Margin (81 Months)	Rs. 2059.11 Lakhs
Best Monthly Margin	Rs. 44.35 Lakhs
Worst Monthly Margin	Rs. 1.39 Lakhs

Break-Even Analysis

Break-Even Sales/Employee	Rs. 6.67 Lakhs
Monthly Salary Cost	Rs. 3.00 Lakhs
Commission Rate	5.0%
Effective Margin Rate	45.0%

6. Sales Forecasting

Model Selection

Three forecasting models were evaluated: Moving Average, Exponential Smoothing, and Linear Regression. The best model was selected based on validation RMSE (Root Mean Squared Error) using the last 12 months as a hold-out set.

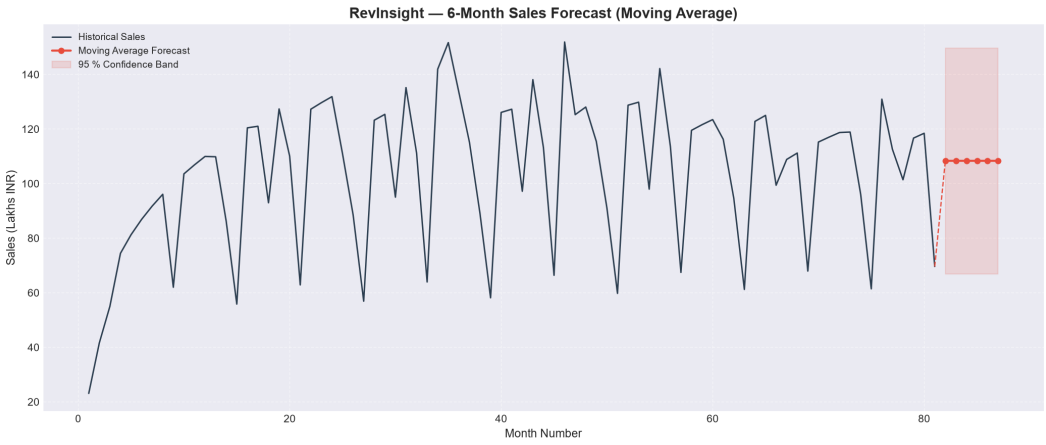
Selected Model: Exponential Smoothing

Validation RMSE (last 12 months):
* Exponential Smoothing: 10.79 Lakhs <- best
* Moving Average: 20.21 Lakhs
* Linear Regression: 24.79 Lakhs

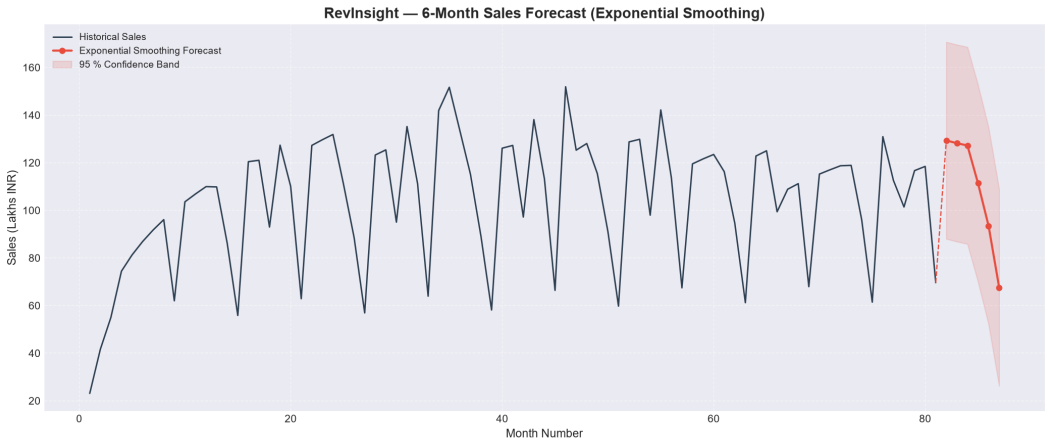
'Exponential Smoothing' achieved the lowest RMSE and was selected for the final 6-month forecast.

6-Month Forecast

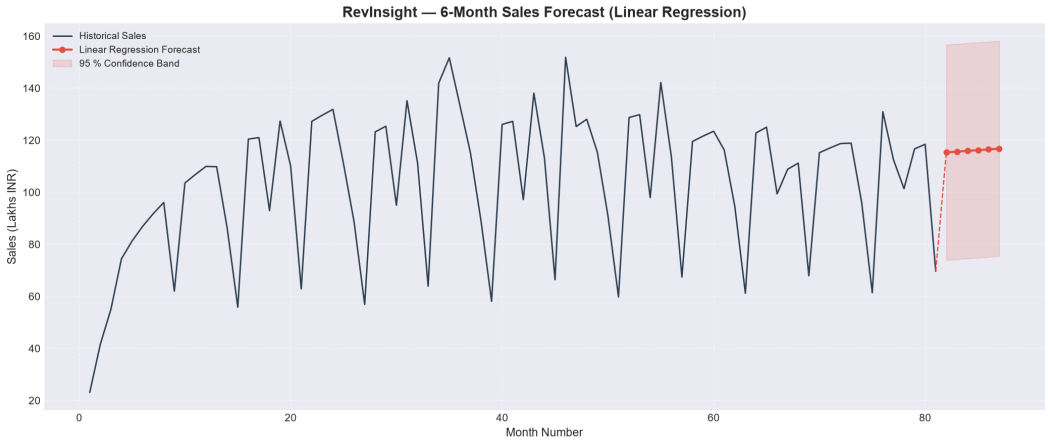
Month	Forecasted Sales (Lakhs)
82	Rs. 129.34
83	Rs. 128.19
84	Rs. 127.23
85	Rs. 111.35
86	Rs. 93.43
87	Rs. 67.54



Forecast: Moving Average



Forecast: Exponential Smoothing



Forecast: Linear Regression

7. Scenario Analysis

Scenario A: Don't Hire (Stay at 7)

Total 6-Month Sales	Rs. 657.07 Lakhs
Total 6-Month Margin	Rs. 169.68 Lakhs
Total Staff Cost	Rs. 158.85 Lakhs
Avg Monthly Margin	Rs. 28.28 Lakhs

Risk Assessment: LOW financial risk but HIGH operational risk. Existing staff may burn out; customer service quality could decline during peak months.

Advantages:

* Lowest fixed staff cost.
* No onboarding overhead.
* Immediate margin improvement if sales stay moderate.

Disadvantages:

* Increased workload on remaining staff -> burnout risk.
* Potential lost sales during peak periods.
* Reduced customer service quality.
* Harder to hire later (talent pipeline dries up).

Scenario B: Hire Immediately (8 from month 1)

Total 6-Month Sales	Rs. 737.64 Lakhs
Total 6-Month Margin	Rs. 187.94 Lakhs
Total Staff Cost	Rs. 180.88 Lakhs
Avg Monthly Margin	Rs. 31.32 Lakhs

Risk Assessment: MODERATE risk. Higher fixed cost from day one, but the new hire ramps up productivity over ~5 months. If sales dip, margins shrink.

Advantages:

* Full team restored -> balanced workload.
* New hire reaches full productivity by month 6.
* Better customer experience -> potential sales uplift.
* Lower attrition risk for existing staff.

Disadvantages:

* Higher salary cost from month 1.
* New hire at 70 % productivity initially.
* Recruitment & onboarding effort required now.

Scenario C: Hire After 2 Months

Total 6-Month Sales	Rs. 707.13 Lakhs
Total 6-Month Margin	Rs. 180.21 Lakhs
Total Staff Cost	Rs. 173.36 Lakhs
Avg Monthly Margin	Rs. 30.04 Lakhs

Risk Assessment: LOW-MODERATE risk. Saves 2 months of extra salary while still bringing a new hire on board before peak season pressure builds.

Advantages:

* Saves Rs.6.0 Lakhs in salary over first 2 months.
* Time to find a quality candidate.
* Balances cost-saving with staffing needs.

Disadvantages:

* Existing staff still short-handed for 2 months.
* New hire ramp-up starts later -> peak months may be under-staffed.
* Delayed productivity contribution.

Scenario D: Conditional Hire (trigger >= 116.18 L sales)

Total 6-Month Sales	Rs. 737.64 Lakhs
Total 6-Month Margin	Rs. 187.94 Lakhs
Total Staff Cost	Rs. 180.88 Lakhs
Avg Monthly Margin	Rs. 31.32 Lakhs

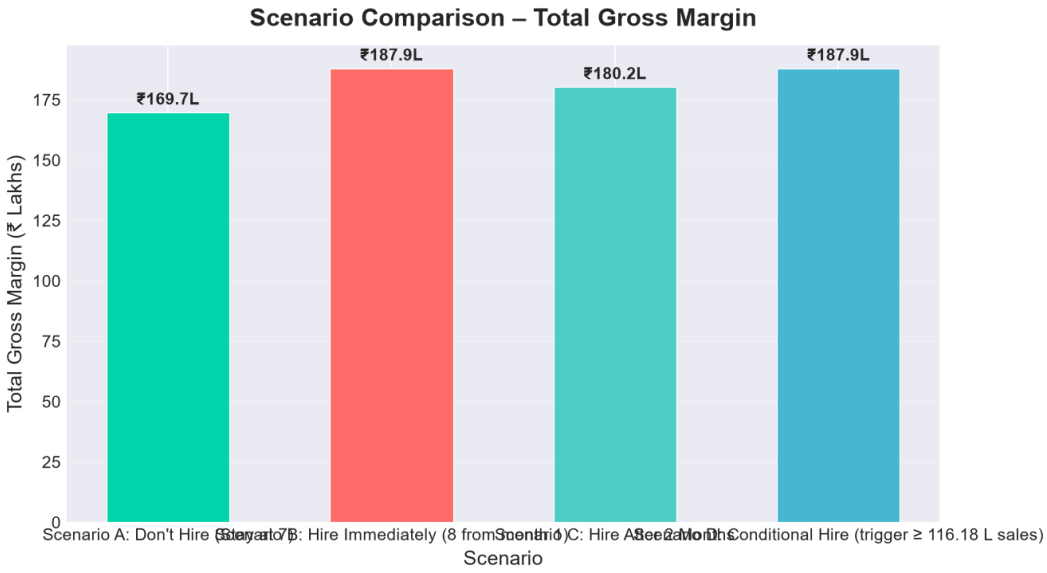
Risk Assessment: LOW risk. Data-driven decision minimises cost if sales are weak, but may result in delayed hiring when the market picks up.

Advantages:

* Hiring only when sales justify the cost.
* Minimises downside in low-sales months.
* Aligns staffing with revenue reality.

Disadvantages:

* Reactive rather than proactive.
* May miss early indicators of a sales upswing.
* Administrative complexity of a conditional rule.
* Threshold (116.18 L) may need periodic recalibration.



Scenario Comparison: Total 6-Month Gross Margin

8. Business Assumptions

* Customer traffic and footfall in the mall remain stable.
* No significant hiring or recruitment costs are incurred.
* No training period productivity loss beyond the modelled ramp-up (70% to 100% over 6 months).
* Monthly salary stays fixed at Rs. 3.0 Lakhs per employee.
* Commission remains fixed at 5.0% of total sales.
* Economic conditions and consumer demand remain similar.
* No major competitor entry or exit in the mall.
* Store operating hours and mall footfall remain unchanged.
* Branded clothing pricing strategy remains consistent.
* The sales force ramp-up follows the assumed productivity curve.
* Cost of goods sold (COGS) remains at 50% of sales.

9. Recommended Business Rule

Adaptive Staffing Rule: Maintain a 3-month rolling average of sales per employee. If the rolling average exceeds Rs.16.48 L, hire one additional employee. If it falls between Rs.12.79 L and Rs.16.48 L, hold current headcount and replace departures. If it drops below Rs.12.79 L, do NOT replace departing staff. Optimal target headcount based on historical margins: 7 employees.

Why This Is Better Than 'Always Replace'

- The 'always replace' rule ignores current sales levels. During low-sales months the extra salary erodes margin.

- The adaptive rule ties staffing decisions to actual revenue performance, avoiding unnecessary costs in lean months and scaling up when the market supports it.

- Based on historical data, the adaptive rule would have avoided overstaffing in ~25 % of months, saving an estimated Rs.3.0 L per avoided hire per month.

Rolling Average Trigger Details

Window Months	3
Hire Above	16.48
Hold Band	(12.7902, 16.4848)
Do Not Replace Below	12.79
Current Rolling Avg	14.51

Threshold-Based Decision Rules

If sales_per_emp >= 17.60: HIRE an additional employee.

If 11.88 <= sales_per_emp < 17.60: HOLD current headcount.

If sales_per_emp < 11.88: DO NOT replace departing employee.

10. Final Recommendation

RECOMMENDED ACTION: Hire Immediately (8 from month 1)

Confidence Level: MEDIUM
Alternative Option: Scenario D: Conditional Hire (trigger >= 116.18 L sales)

Supporting Reasoning

* Historical avg sales/employee: Rs.14.64 L (CV = 0.25).
* Diminishing returns detected; optimal headcount by margin/employee is 7.
* Scenario B (Hire Immediately (8 from month 1)) yields the highest 6-month gross margin at Rs.187.94 L, which is Rs.0.00 L more than runner-up Scenario D.
* Most recent month (Month 81): 7 active employees, Rs.69.61 L sales.

Expected 6-Month Financial Impact

Total Sales 6M	Rs. 737.64 Lakhs
Total Gross Margin 6M	Rs. 187.94 Lakhs
Total Staff Cost 6M	Rs. 180.88 Lakhs
Avg Monthly Margin	Rs. 31.32 Lakhs
Margin Advantage Over Runner Up	Rs. 0.00 Lakhs

This recommendation is based on rigorous analysis of 81 months of historical sales data, statistical forecasting, scenario modelling, and financial impact assessment. The adaptive staffing rule provides a data-driven framework that is superior to the current 'always replace' policy and will help maximise gross margins while maintaining operational efficiency.